

Synthetic Biology Outreach Toolkit

Lesson Menu: nine ready-to-run, hands-on lessons for grades 3-6

Bring them into any classroom · cheap, portable materials

Free to use & adapt · CC0 (public domain)

● Foundations (read DNA)

● Engineering (edit & build)

● Application (solve a problem)

1 Intro: What Is Synthetic Biology?

Grades 3-6 · 10-15 min · \$0

A discussion opener on what DNA and synthetic biology are, using real-world examples kids recognize.

2 Build-a-DNA Double Helix

Grades 3-6 · 25-35 min · \$10-20

Build a candy or bead double helix and discover the A-T / C-G base-pairing rule.

3 Name-in-DNA Bracelets

Grades 3-5 · 25-35 min · \$15-25

Spell your name in a three-bead code and take home a bracelet - an intro to codons.

4 Strawberry DNA Extraction

Grades 4-6 · 30-40 min · \$15-20

Pull real, visible DNA from a strawberry using soap, salt, and rubbing alcohol.

5 Monster Genetics

Grades 3-5 · 30-40 min · \$5-10

Decode a DNA recipe into a monster, then swap a gene with a partner to transform it.

6 CRISPR Cut-and-Paste

Grades 5-6 · 30-40 min · \$2-5

Be the molecular scissors: find a target DNA sequence, cut it, and paste in a new one.

7 Engineer a Glowing Microbe

Grades 3-6 · 30-40 min · \$10-15

Insert a jellyfish glow gene into a paper microbe and design a useful new job for it.

8 Clean Water: Sensor & Filter

Grades 5-6 · 45-60 min · \$15-25

Debate contamination, see how a biosensor detects a toxin, then design and test a filter.

9 Design-a-Synbio-Solution

Grades 4-6 · 30-40 min · \$2-5

Capstone: combine biological parts (chassis + sensor + output) to solve a real problem.